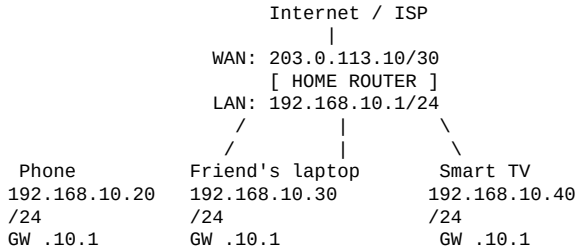
Network Routing and OSPF Practical

Detailed answers to Questions 1–5

Note: All IP addresses, routers, cities and company branches below are fictional examples for learning. Replace interface names with those on your own equipment. Do not apply router commands to a live network unless you have permission and a backup.

1. Network Diagram with Three Devices and Router Interfaces

Example home LAN: the router's LAN interface is 192.168.10.1/24. The phone, friend's laptop and smart TV use unique addresses from the same subnet. The WAN interface connects to the internet provider and receives an example address from the ISP.



Address labels: Router LAN/gateway = 192.168.10.1/24; phone = 192.168.10.20/24; laptop = 192.168.10.30/24; smart TV = 192.168.10.40/24. Default gateway on all three devices is 192.168.10.1. The WAN example 203.0.113.10/30 is from a documentation-only range, not a usable public ISP address.

Because all three devices are in 192.168.10.0/24, they can communicate locally through the LAN/switching function of the router, subject to Wi-Fi isolation and firewall settings. Traffic going to a different subnet is sent to the default gateway.

2. How a Packet Travels to a Different Subnet

Scenario: Laptop A = 192.168.10.30/24; destination friend's PC = 192.168.20.50/24. Router R1 connects the home LAN to a transit network, and Router R2 connects the transit network to the friend's LAN.

```
Laptop A                R1                    R2              Friend PC
192.168.10.30 -> 192.168.10.1 -> 10.0.0.2 -> 192.168.20.1 -> 192.168.20.50
 LAN /24       Home router    transit link    Friend router       LAN /24
```

Step-by-step:

1. Laptop A compares its own subnet (192.168.10.0/24) with the destination (192.168.20.50). The destination is outside its local subnet.

2. The laptop consults its routing table. A typical entry is 192.168.10.0/24 as directly connected and 0.0.0.0/0 via 192.168.10.1 as the default route.

3. The laptop sends the Ethernet/Wi-Fi frame to the router's LAN MAC address. The IP packet still has source 192.168.10.30 and destination 192.168.20.50.

4. R1 checks its routing table using longest-prefix match. If it has a route to 192.168.20.0/24 via next hop 10.0.0.2, it forwards the packet to R2 through the transit interface.

5. R2 checks its table. Since 192.168.20.0/24 is directly connected, R2 sends the packet on the friend's LAN and resolves the destination PC's MAC address with ARP.

6. The friend's PC receives the packet. The reply travels back using its own default gateway and valid return routes. A missing return route or firewall rule can prevent a successful connection.

Example routing-table entries

Router	Destination/prefix	Next hop	Interface
R1	192.168.10.0/24	Directly connected	G0/0
R1	10.0.0.0/30	Directly connected	G0/1
R1	192.168.20.0/24	10.0.0.2	G0/1
R2	10.0.0.0/30	Directly connected	G0/1
R2	192.168.20.0/24	Directly connected	G0/0
R2	192.168.10.0/24	10.0.0.1	G0/1

3. Fictional Zomato-Style Delivery Network Routing Table

Assume a fictional food-delivery company has three branch LANs in Ahmedabad, Mumbai and Pune. Each branch uses a different /24 subnet. The routers are linked by a private WAN. These names and addresses are invented for this assignment.

Branch / destination	Network address	Subnet mask	Next hop	Interface
Ahmedabad LAN	10.10.1.0	255.255.255.0 (/24)	Directly connected	G0/0
Mumbai LAN	10.10.2.0	255.255.255.0 (/24)	10.0.0.2	G0/1
Pune LAN	10.10.3.0	255.255.255.0 (/24)	10.0.0.6	G0/2

Routing interpretation: The Ahmedabad router can reach its own LAN directly. For the Mumbai subnet, it forwards traffic to next hop 10.0.0.2 over the WAN interface; for Pune, it forwards to 10.0.0.6. In an actual deployment, next-hop addresses must be reachable on the specified interface, and routes must exist in both directions.

Illustrative route commands (Cisco IOS-style):

```
! On Ahmedabad router (example only)
ip route 10.10.2.0 255.255.255.0 10.0.0.2
ip route 10.10.3.0 255.255.255.0 10.0.0.6
```

The first command adds a static route to the Mumbai LAN through 10.0.0.2. The second adds a static route to the Pune LAN through 10.0.0.6. The actual interface/next-hop design depends on the WAN topology.

4. Static Routing vs Dynamic Routing (RIP/OSPF)

Static routing means an administrator manually enters routes. Dynamic routing protocols let routers exchange route information and recalculate routes as the network changes.

Method	3 advantages	3 disadvantages	Example
Static routing	1. Simple for small networks. 2. No routing-protocol update traffic. 3. Predictable; administrator controls the path.	1. Manual updates are needed after topology changes. 2. Does not automatically reroute around a failed link. 3. Hard to maintain across many branches.	A small shop's single backup link or a fixed route to a branch network.
Dynamic routing (RIP/OSPF)	1. Learns routes from other routers. 2. Can recalculate after a link or route changes. 3. Scales better across many interconnected branches.	1. More complex to configure and troubleshoot. 2. Uses CPU, memory and bandwidth for protocol operation. 3. Misconfiguration can spread bad routes or cause instability.	A delivery company with multiple city branches can use OSPF to share routes and adapt if a WAN link fails.

Real-world app connection: A food-delivery app relies on servers, databases and network links to process orders. Dynamic routing does not choose a delivery rider's road route; instead, it helps the underlying IP network find paths between branch routers and services. RIP is a simpler distance-vector protocol with limited scalability; OSPF is a link-state protocol commonly better suited to larger enterprise networks.

5. Sample OSPF Configuration Between Two Routers

Topology: R1 LAN = 192.168.1.0/24; R2 LAN = 192.168.2.0/24; inter-router link = 10.0.0.0/30. R1 uses 10.0.0.1/30 and R2 uses 10.0.0.2/30. Interface names below are examples and must match the actual router.

```
R1 (Cisco IOS-style)
interface GigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
exit
interface GigabitEthernet0/1
 ip address 10.0.0.1 255.255.255.252
 no shutdown
exit
router ospf 1
 router-id 1.1.1.1
 network 192.168.1.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.3 area 0
```

Line-by-line explanation for R1:

• `interface GigabitEthernet0/0` selects the LAN-facing interface.

• `ip address 192.168.1.1 255.255.255.0` assigns the router's LAN address and subnet mask.

• `no shutdown` enables the interface administratively.

• `interface GigabitEthernet0/1` selects the link to R2; the following address puts it in the 10.0.0.0/30 transit subnet.

• `router ospf 1` starts an OSPF process with local process ID 1. The process ID is locally significant on Cisco IOS.

• `router-id 1.1.1.1` sets a unique OSPF router ID for R1.

• `network ... area 0` enables OSPF on matching interfaces and advertises those connected networks into OSPF area 0. The numbers after the network are wildcard masks, not subnet masks.

R2 configuration

```
R2 (Cisco IOS-style)
interface GigabitEthernet0/0
 ip address 192.168.2.1 255.255.255.0
 no shutdown
exit
interface GigabitEthernet0/1
 ip address 10.0.0.2 255.255.255.252
 no shutdown
exit
router ospf 1
 router-id 2.2.2.2
 network 192.168.2.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.3 area 0
```

R2's LAN uses 192.168.2.1/24; its WAN/transit interface is 10.0.0.2/30. Router ID 2.2.2.2 must be different from R1's. Both routers put the inter-router link in area 0 and advertise their own LAN subnet.

Verification steps

1. Confirm interfaces are up with `show ip interface brief`.

2. Check OSPF neighbors with `show ip ospf neighbor`. The routers should form a neighbor relationship when the link, area, timers, and OSPF settings are compatible.

3. Inspect learned routes using `show ip route ospf` or `show ip route`.

4. From a PC in 192.168.1.0/24, ping a reachable host in 192.168.2.0/24. Ensure hosts have correct default gateways and firewalls permit ping.

5. If no adjacency forms, check cabling, interface status, IP/mask, matching area, duplicate router IDs, and whether OSPF is enabled on the transit interface.

Important: This is a sample Cisco IOS-style configuration. Packet Tracer device models and router operating systems may vary. Use the actual interface names shown by your device, and save configuration only after verifying the lab works.